Mock 11: Wrong-Answer Notes

CFA Level II · 4 wrong answers, one per page

40 / 44

Mock 11 (90.9%)

Score by topic

Topic	Score
Derivatives	2/4
Financial Statement Analysis	3/4
Equity Valuation	7/8
Economics	4/4
Corporate Issuers	4/4
Portfolio Management	4/4
Fixed Income	4/4
Ethical and Professional Standards	8/8
Quantitative Methods	4/4

Questions at a glance

Q	Topic	You	Ans
Q23	Equity	A	B
Q30	Deriv	C	B
Q31	Deriv	B	A
Q39	FSA	A	B

Mock 11 Q23 **PEG ratio: forward P/E (price ÷ NEXT-four-quarter EPS) ÷ growth rate in %**

You **A** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Alves directs Ramos to use a valuation metric that would allow for a meaningful ranking of relative value of the three companies' shares. Exhibit 1 provides selected financial information for the three companies.

Exhibit 1: Selected Financial Information for Norte, Costa, and Valle (Mexican Peso, MXN)

	Norte	Costa	Valle
Stock's current price	18.60	64.80	118.30
Diluted EPS (last four quarters)	0.95	3.35	-0.12
Diluted EPS (next four quarters)	1.05	2.70	2.35

Exhibit 2: Earnings Growth Rate Estimates over Five Years

Company	Earnings Growth Rate Estimate (%)
Norte	15.40
Costa	24.50
Valle	28.60

Question 23. Based on Exhibit 1 and Exhibit 2, which company's shares are the most attractively priced based on the five-year forward P/E-to-growth (PEG) ratio?

OPTIONS

A	Norte	✗ Your pick	Your pick: Norte has a low forward P/E (17.7) but also the lowest growth (15.40%), so its PEG (1.15) is above Costa's.
B	Costa	✓ Correct	Correct: Costa has the lowest forward PEG (0.98).
C	Valle		Valle has the highest forward PEG (1.76): a high forward P/E (50.3) is not paid for by its growth, even though its growth is the highest.

Trigger "five-year forward P/E-to-growth (PEG) ratio"
"Diluted EPS (next four quarters)"
"Earnings Growth Rate Estimate (%)"

Rule next time: Forward PEG = (price ÷ EPS of the next four quarters) ÷ growth rate (as a number, 15.40 not 0.154). The LOWEST PEG is the cheapest per unit of growth.

FORMULA / RULE

Forward P/E = price ÷ forward EPS
PEG = forward P/E ÷ (growth rate × 100)
Lowest PEG = most attractively priced

WHY THE ANSWER IS B

1. Norte: 18.60 ÷ 1.05 = 17.71; ÷ 15.40 = 1.15.
2. Costa: 64.80 ÷ 2.70 = 24.00; ÷ 24.50 = 0.98.
3. Valle: 118.30 ÷ 2.35 = 50.34; ÷ 28.60 = 1.76: **B**.

Apply to other questions: "Forward" → next-four-quarter EPS; compute all three companies before choosing; a low P/E alone is not enough, divide by the growth.

Mock 11 Q30 **FRA value after initiation: new FRA rate from MRRs, then discount to TODAY over the days to the END of the underlying period**

You **C** · Answer **B**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

One month ago (30 days), Wells entered a pay-fixed 3 × 6 forward rate agreement (FRA) at a rate of 2.40% with a notional amount of US$5,000,000. Now, 30 days after entering the FRA, the two-month MRR is 1.5% and the five-month MRR is 2.5%.

Question 30. The current value of Wells's FRA is closest to:

OPTIONS

A	US$9,485.		Leaves the settlement undiscounted (9,485).
B	US$9,387.	✓ Correct	Correct: discounted over 150 days = US$9,387.
C	US$9,426.	✗ Your pick	Your pick: discounts over 90 days (the length of the underlying period) instead of 150 days (time from today to its end): US$9,426.

Trigger “pay-fixed 3 × 6 forward rate agreement (FRA) at a rate of 2.40%”
“30 days after entering the FRA”
“the two-month MRR is 1.5% and the five-month MRR is 2.5%”

Rule next time: After 30 days the FRA has 2 months left and the underlying 3-month period ends in 5 months (150 days). Find the new 2 × 5 rate, take the difference with the contract rate, and discount over 150 days (the end of the underlying period), not 90.

FORMULA / RULE

New FRA rate = [(1 + L_long × t_long/360) ÷ (1 + L_short × t_short/360) − 1] × 360/(t_long − t_short)
Value (pay-fixed) = (new rate − contract rate) × (days/360) × notional ÷ (1 + L_long × t_long/360)

WHY THE ANSWER IS B

1. New rate = [(1 + 0.025 × 150/360) ÷ (1 + 0.015 × 60/360) − 1] × 360/90 = 3.159%.
2. Difference: (3.159% − 2.40%) × 90/360 × 5,000,000 = 9,485.
3. Discount: 9,485 ÷ (1 + 0.025 × 150/360) = US$9,387: **B**.

Apply to other questions: FRA value: the discount period is from today to the END of the underlying period, i.e. the longer MRR (here 150 days), never the length of the period itself.

Mock 11 Q31 **Currency swap fixed rate: use the PV factors of the leg you PAY; sum of all factors in the denominator**

You **B** · Answer **A**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Rao, who is based in Hong Kong SAR, requires a one-year bridge loan in euros to fund operations in Germany. He wants to fund this loan at a competitive rate. Fischer advises Rao to borrow HK dollars domestically and then enter into a one-year foreign currency swap to swap into euros. Exhibit 1 below provides Hong Kong and euro spot interest rates and present value factors.

Exhibit 1: Hong Kong and Euro Spot Interest Rates

Days to Maturity	HK Dollars Spot Interest Rates (%)	HK Dollars Present Value Factors	Euro Spot Interest Rates (%)	Euro Present Value Factors
90	0.620	0.9985	0.355	0.9991
180	0.780	0.9961	0.405	0.9980
270	0.860	0.9936	0.440	0.9967
360	0.950	0.9906	0.470	0.9953

Question 31. Based on the information in Exhibit 1, the annual fixed swap rate Rao would pay is closest to:

OPTIONS

A	0.47%.	✓ Correct	Correct: the euro fixed rate Rao pays, 0.47%.
B	0.95%.	✗ Your pick	Your pick: 0.95% is the HK dollar fixed rate, which Rao RECEIVES.
C	1.89%.		Divides by the last factor instead of the sum of the factors (1.89%).

Trigger “requires a one-year bridge loan in euros”
“borrow HK dollars domestically and then enter into a one-year foreign currency swap to swap into euros”
“Euro Present Value Factors”

Rule next time: Rao needs euros, so in the swap he receives HK dollars (to pay his HK dollar loan) and PAYS euros: use the EURO factors. Fixed rate per period = (1 − last PV factor) ÷ sum of the PV factors; multiply by 4 for the annual rate.

FORMULA / RULE

Fixed rate per period = (1 − PV_n) ÷ Σ PV_i
Annual rate = per-period rate × periods per year (quarterly: × 4)

WHY THE ANSWER IS A

1. Euro leg: (1 − 0.9953) ÷ (0.9991 + 0.9980 + 0.9967 + 0.9953) = 0.1178%.
2. × 4 = 0.47%: **A**.
3. HK dollar leg: (1 − 0.9906) ÷ 3.9788 × 4 = 0.95% (the leg Rao receives).

Apply to other questions: Currency swap: decide which currency you PAY first, then take that currency's factors. The borrower ends up paying the currency he needs.

Mock 11 Q39 **Equity method carrying value: cost + share of net income − share of dividends − amortisation of the fair value excess**

You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Investment in Calder Co.

On 1 January 2016, Veridian acquired a 30% equity interest in Calder Co. for $620 million. Exhibit 2 presents selected financial information for Calder on the acquisition date. Thornton notes that the plant and equipment are depreciated on a straight-line basis and have 10 years of remaining life.

Exhibit 2: Selected Financial Data for Calder Co., 1 January 2016 (Acquisition Date) ($ millions)

	Book Value	Fair Value
Current assets	300	300
Plant and equipment	3,100	3,400
Total assets	3,400	3,700
Liabilities	1,900	1,900
Net assets	1,500	1,800

Thornton notes that, for fiscal year 2016, Calder reported net income of $400 million and paid dividends of $240 million.

Question 39. Using only the information from Exhibit 2 and the 2016 results, the carrying value of Veridian's investment in Calder at the end of 2016 is closest to:

OPTIONS

A	$668 million.	✗ Your pick	Your pick: leaves out the amortisation of the plant excess (668).
B	$659 million.	✓ Correct	Correct: 620 + 120 − 72 − 9 = $659 million.
C	$731 million.		Does not deduct the dividends received (731).

Trigger “depreciated on a straight-line basis and have 10 years of remaining life”
“Plant and equipment | 3,100 | 3,400”
“reported net income of $400 million and paid dividends of $240 million”

Rule next time: Fair value above book value on depreciable plant creates an excess that must be amortised: 30% × (3,400 − 3,100) ÷ 10 years = 9 per year. Carrying value = cost + 30% × net income − 30% × dividends − 9.

FORMULA / RULE

> Investment = cost + % × net income − % × dividends − amortisation
> Amortisation = % × (FV − BV) ÷ remaining life (for the part of the excess on depreciable assets)

WHY THE ANSWER IS B

1. Share of net income: 0.30 × 400 = 120; dividends: 0.30 × 240 = 72.
2. Amortisation: 0.30 × (3,400 − 3,100) ÷ 10 = 9.
3. 620 + 120 − 72 − 9 = **659**: **B**.

Apply to other questions: Equity method: dividends reduce the investment; and check whether FV differs from BV on depreciable assets (amortise) or on land (no amortisation).